

Why a screen feels good, decomposed into properties you can instrument

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CITATION GATE

Cleared 2026-09-04. Every citation was checked against a primary source. The unresolved complexity-order attribution was re-derived from scratch as Van Geert & Wagemans (2020); the phantom "Post & Nadal" attribution is retired. A remaining **LIKELY** grades the strength of the inference from a real source to the interface claim, never the existence of the source.

ABSTRACT

"This interface feels premium" is treated as ineffable taste. It is not. Aesthetic response decomposes into a small set of mechanisms, each tied to a **measurable visual property** — and most of those properties are already instrumentable. This paper maps eight mechanisms from experimental aesthetics and reward neuroscience onto the properties an interface exposes (contrast, symmetry, curvature, colour coordinates, complexity, and a behavioral liking/wanting signature), grades each by evidence, and names the folklore to exclude (the golden ratio, the Mehrabian 7-38-55 misapplication, rigid colour-psychology universals). The organizing claim, and the reason this sits in one program with the Cognitive Attack Surface: the single mechanism that does *not* reduce to a static property — the dissociation of **liking** from **wanting** — is exactly what separates good design (liking and wanting aligned) from a dark pattern (wanting engineered past liking). Aesthetic quality and manipulation are measured on the same axis.

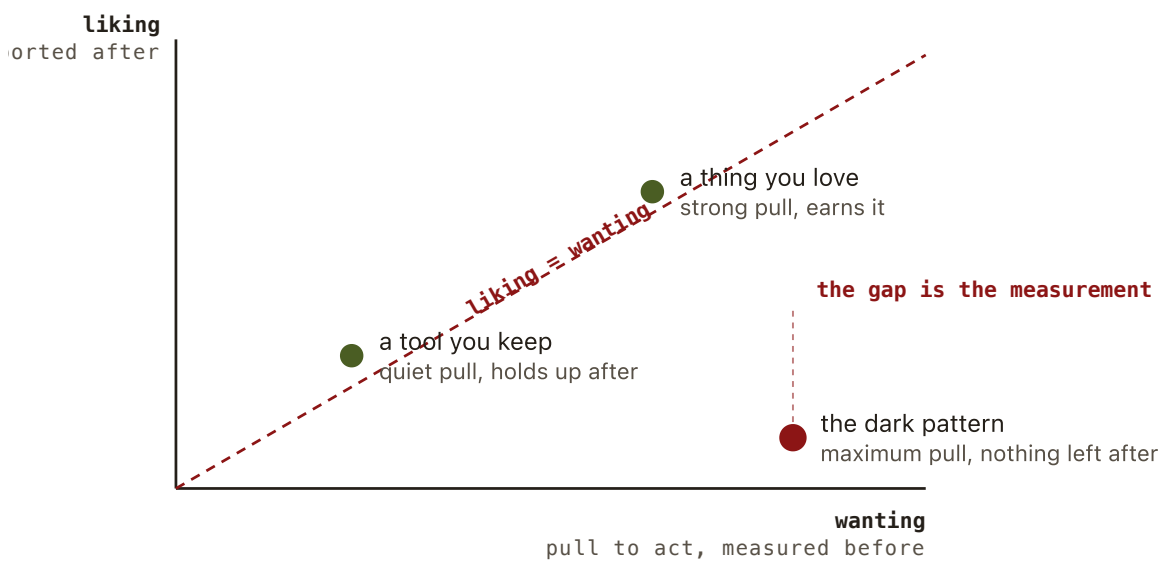
1. The thesis: felt quality is a readout of measurable properties

Aesthetic pleasure is, in large part, a readout of how easily the perceptual system processes a stimulus, misattributed to the stimulus as "beauty" (**processing fluency**: Reber, Schwarz & Winkielman 2004, *PSPR* — **VERIFIED**; one of the most replicated findings in experimental aesthetics). Fluent stimuli — high contrast, clear figure-ground, symmetry, prototypicality, prior exposure — are liked more, and even raise

smiling-muscle activation. This is the load-bearing premise: the properties that make a screen easy to process are the properties that make it *feel good*, and those properties are measurable.

The structure of the response is the **aesthetic triad** (Chatterjee & Vartanian 2014, *TICS* — **LIKELY**, as an organizing framework): sensory-motor processing, emotion-valuation, and meaning-knowledge interact to produce aesthetic experience, unfolding across stages (Leder et al. 2004, used explicitly to frame web/product-design studies). The triad is the paper's scaffold, not an instrument — sensory-motor maps to §§2,5,6,7; emotion-valuation to §3; meaning-knowledge to prototypicality/familiarity in §2 and §4.

2. The mechanisms and what each one measures



Berridge & Kringelbach 2015 · Salimpoor et al. 2011

FIGURE 1 – THE AXIS THE WHOLE PROGRAM IS READ ON

Reward splits into hedonic liking and incentive wanting, and the two dissociate. Honest design sits near the diagonal. A dark pattern is engineered downward and to the right: it maximises the pull to act while the reported experience afterward falls away. Aesthetic quality and manipulation are the same measurement taken in opposite directions, which is why one instrument covers both.

Processing fluency (**VERIFIED**) → contrast, figure-ground clarity, symmetry, prototypicality, exposure/familiarity. The umbrella mechanism under which most of the others sit.

Liking vs wanting (**VERIFIED** mechanism; **LIKELY** for the aesthetic application). Reward has two dissociable components — hedonic *liking* (opioid/endocannabinoid hedonic hotspots) and incentive *wanting* (dopaminergic pull) — that come apart (Berridge & Kringelbach 2015, *Neuron*). The aesthetic bridge is the music-reward literature: dopamine release during *anticipation* of a musical peak is

anatomically distinct from the peak-pleasure itself (Salimpoor et al. 2011, *Nat. Neurosci.* — VERIFIED); the strongest bridge from food/drug reward into an aesthetic domain, and anatomically explicit: caudate during anticipation, nucleus accumbens at the peak). This is the one mechanism that maps not to a static pixel property but to a **behavioral/temporal signature**: does a UI element pull an action (tap, scroll, buy) disproportionate to how much the user reports liking it afterward? Instrument it as the gap between anticipatory pull and post-outcome affect (dwell-then-regret, action-vs-reported-liking, or pupillometry/facial-EMG if available). **This is the axis that unifies the program**: good design keeps liking and wanting aligned; a dark pattern engineers wanting past liking. Aesthetic quality and manipulation are the same measurement, read in opposite directions.

Complexity / arousal — the inverted-U (VERIFIED as a classical claim, CONTESTED in modern replication, and honestly noisy). Preference peaks at moderate complexity (Berlyne 1971, *Aesthetics and Psychobiology* — the collative variables of novelty, uncertainty, conflict and complexity, and the inverted-U between complexity and hedonic tone; Wundt's curve). The modern review qualifies it: order and complexity interact rather than acting as one axis, and high-order/high-complexity images read as *fascinating* while high-order/low-complexity images read as *soothing* (Van Geert & Wagemans 2020 — VERIFIED). A single complexity number therefore cannot predict preference on its own, which is the honest reason this mechanism sits lowest in the build order, but complexity effects are confounded with **prototypicality** — goodness-of-fit to a category often explains what looks like a complexity effect (Martindale & Moore 1988). Measurable: visual entropy, edge density, element count, and distance-from-category-prototype. State the messiness; do not present the inverted-U as a clean law.

Colour preference (VERIFIED mechanism / CONTESTED universal content). Ecological valence theory: preference for a colour tracks the average valence of the things people associate with it (Palmer & Schloss 2010, *PNAS*) — which is exactly *why* colour meaning is culture- and experience-dependent, not fixed. Measurable, and already fully instrumented in this program: CIELAB/OKLCH coordinates (L, C, h°), chroma, lightness — and chroma read *in context*, not as an isolated swatch.

Contour / curvature (VERIFIED). Sharp/angular contours carry a mild threat signature and are liked less; curves are preferred (Bar & Neta 2006, *Psych. Science*; extended to architecture via fMRI by Vartanian et al. 2013, *PNAS*). Load-bearing caveat that ties back to liking/wanting: curvature buys **liking** (beauty ratings) but did *not* significantly move **approach-avoidance behavior** — curves make a thing more beautiful without necessarily making it more engaged-with. Measurable: corner radius, average radius of curvature along an edge — trivially parsed from CSS border-radius and SVG path data.

Contrast / legibility (VERIFIED vision science; LIKELY for APCA as *the* instrument). Legibility is an aesthetic and cognitive-load variable, not just an accessibility checkbox — insufficient contrast is a disfluency cost. APCA (the perceptual-contrast successor to WCAG 2's ratio math, which false-passes in dark mode) is the best available instrument, still maturing through WCAG 3. Measurable, already instrumented: APCA Lc.

Symmetry (**VERIFIED**) → a bilateral/radial symmetry score over element layout. A fluency cue and a fitness/order cue; robust across abstract patterns, not just faces (Jacobsen & Höfel 2002 — **VERIFIED**; symmetry was the highest correlate of judged beauty and complexity the second, with substantial individual differences. Enquist & Arak 1994 — **VERIFIED**; symmetry preference arises as a by-product of position- and orientation-invariant recognition rather than as a signalling adaptation).

3. Quarantine — the folklore an interface-aesthetics paper must exclude

Naming these is a credibility feature, because design writing repeats all of them:

- **The golden ratio as a beauty law** — **CONTESTED** /debunked-as-stated. Retrofitted onto artifacts after the fact, not predictive; the Pacioli–golden-ratio link is itself a documented 1799 historical error; Fechner-rectangle replications are inconsistent (Markowsky 1992 is the canonical debunk). Excluded from the instrument set.
- **Mehrabian's 7-38-55** — **DEBUNKED** as generally applied. The 1967 studies measured only attitude/liking communication under verbal-nonverbal *inconsistency*, single spoken words; never a law that "communication is 93% nonverbal."
- **Rigid colour-psychology universals** ("blue = trust," "red = urgency" cross-culturally fixed) — **CONTESTED**. Cross-country divergence is the dominant finding (purple↔anger only in Poland; yellow↔jealousy only in Germany). Ecological valence theory *predicts* this variation — so EVT argues against the rigid-universal version. Any colour-meaning claim must name its population.
- **F-pattern / Z-pattern as a universal reading law** — **CONTESTED** (verified 2026-09-04). The F-pattern is a real 2006 eyetracking observation from Nielsen Norman, and NN/g itself has since published a clarification that scanning does not always take an F and that the finding is widely misread as a law. An independent analysis across 99 sites reported no evidence of an F-shaped bias at all. Treat it as one observed pattern on text-heavy pages, never as a scanning law, and never as an instrument.

4. The mechanism → measurable-property map (the core deliverable)

MECHANISM	GRADE	MEASURABLE PROPERTY	INSTRUMENT STATUS
Processing fluency	VERIFIED	contrast · clarity · symmetry · prototypicality · exposure	contrast: HAVE (APCA); symmetry/prototypicality: BUILD
Aesthetic triad	LIKELY (frame)	— (organizing scaffold)	—
Liking vs wanting	VERIFIED / LIKELY (aesthetic)	anticipation-vs-consumption gap; action pulled vs liking reported	research base HAVE; in-product metric: BUILD
Complexity inverted-U	VERIFIED (classical) / CONTESTED (modern; order interacts)	visual entropy · edge density · prototype-distance	BUILD (lowest priority — noisiest)
Colour preference (EVT)	VERIFIED / CONTESTED (content)	CIELAB/OKLCH L,C,h° · chroma · lightness	HAVE (colour-grade tool)
Contour / curvature	VERIFIED	corner radius · curvature	BUILD (trivial — parse CSS/SVG)
Contrast / legibility	VERIFIED / LIKELY (APCA)	APCA Lc	HAVE (APCA + lightness)
Symmetry	VERIFIED	bilateral/radial symmetry score	BUILD (with fluency detector)
GOLDEN RATIO	CONTESTED / DEBUNKED	—	QUARANTINE ONLY

The build gap, concretely. Already in hand: colour (CIELAB tool), contrast/legibility (APCA), light exposure (melanopic-dose meter), and the liking/wanting research base. Two instruments would put the whole thesis on her own measurement stack rather than citation alone: (1) a **symmetry + prototypicality detector** (serves fluency, complexity, and symmetry at once — the highest-leverage single build), and (2) a **corner-radius / curvature analyzer** (trivial — parses existing geometry). The complexity/entropy metric is the noisiest mechanism and the lowest priority.

5. Why this is one program, not a side quest

The liking/wanting axis (§2) is the same axis the Cognitive Attack Surface papers measure. There, a technique is manipulative when it drives wanting past liking — the covert, disclosure-failing tier. Here, an interface is *good* when it keeps them aligned, and a *dark pattern* when it doesn't. The neuroaesthetics work and the cognitive-security work are the same instrument pointed at two ends of one continuum: what makes an interface feel good, and what makes it exploit you, are read on the same measured axis. That is the through-line of the whole program — the measurement layer, applied to felt experience.

References and evidence grades

Citation gate run 2026-09-04. Every item previously listed as "verify before citing" was checked against a primary source. No unverified citation remains in this paper.

Verified this pass (were on the v0.1 punch list):

- Berlyne, D. E. (1971). *Aesthetics and Psychobiology*. Appleton-Century-Crofts, New York. — collative variables; inverted-U of complexity and hedonic tone.
- Enquist, M., & Arak, A. (1994). Symmetry, beauty and evolution. *Nature*, 372, 169–172. — symmetry preference as a by-product of position- and orientation-invariant recognition, not a signalling adaptation.
- Jacobsen, T., & Höfel, L. (2002). Aesthetic judgments of novel graphic patterns: analyses of individual judgments. *Perceptual and Motor Skills*, 95, 755–766. — symmetry the highest correlate of judged beauty, complexity second, with substantial individual differences.
- Markowsky, G. (1992). Misconceptions about the golden ratio. *The College Mathematics Journal*, 23(1), 2–19. — the canonical debunk.
- Martindale, C., & Moore, K. (1988). Priming, prototypicality, and preference. *Journal of Experimental Psychology: Human Perception and Performance*, 14, 661–670. — prototypical exemplars preferred; the confound with complexity.
- Salimpoor, V. N., Benovoy, M., Larcher, K., Dagher, A., & Zatorre, R. J. (2011). Anatomically distinct dopamine release during anticipation and experience of peak emotion to music. *Nature Neuroscience*, 14(2), 257–262. — caudate during anticipation, nucleus accumbens at peak; the anatomical basis of the wanting/liking split.
- Van Geert, E., & Wagemans, J. (2020). Order, complexity, and aesthetic appreciation. *Psychology of Aesthetics, Creativity, and the Arts*, 14(2), 135–154. doi:10.1037/aca0000224.

Verified in the v0.1 pass, unchanged: Reber, Schwarz & Winkielman (2004); Chatterjee & Vartanian (2014); Leder et al. (2004); Berridge & Kringelbach (2015); Palmer & Schloss (2010); Bar & Neta (2006); Vartanian et al. (2013); the Mehrabian origin and scope.

Retired. The v0.1 punch list carried three entries that this paper never actually cites in its body: Nadal et al. (2010), Silvia & Barona (2009), and a "Post & Nadal" attribution that resolves to no real paper. The

complexity-order claim they were standing in for is now carried by Van Geert & Wagemans (2020), derived from scratch. Nothing in this paper depends on the retired entries.

Quarantined, cited only as folklore to exclude: the golden ratio as a beauty law; the Mehrabian 7-38-55 figures applied outside their original scope; rigid colour-psychology universals.

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Every citation graded on the page. Sources that could not be verified are printed as unverified.